Economic Fibrillation — Crisis-Window Synchronization (REAL GDP)

Robustness run on constant-price data: Maddison Project Database 2023, real GDP per capita (inflation- and PPP-adjusted, constant international dollars, 26 economies, to 2022)

1 Executive summary

This repeats the crisis-window analysis using **real (constant-price) GDP** instead of current US dollars, to check that the findings are not an artefact of inflation or exchange-rate movement. They are not — the core result is confirmed and, if anything, **stronger**. Synchronization in global-crisis years averages r = 0.70, against r = 0.52 in normal years and only r = 0.30 in regional crises, and the relative test is more significant than before (p = 0.020). The two most recent global crises are now unmistakable synchronization *peaks*: the Global Financial Crisis reaches r = 0.93 (92th percentile) and COVID-19 reaches r = 0.98 (the single most synchronized year in the record), followed by a dramatic collapse to r = 0.16 by 2022 — the clearest 'fibrillation' signature in either dataset.

2 Data and method

Method is identical to the current-dollar analysis: ln(real GDP per capita) → Butterworth band-pass (≈ 2–9 yr, order 2) → Hilbert phase → Kuramoto order parameter r(t), for N = 26 economies over 1970–2022. The only change is the input series. Real GDP per capita removes inflation and exchange-rate movement, and (dividing by population) also the slow demographic trend — leaving a cleaner business cycle. Source: Maddison Project Database 2023 (Bolt & van Zanden, 2024).

3 Overview

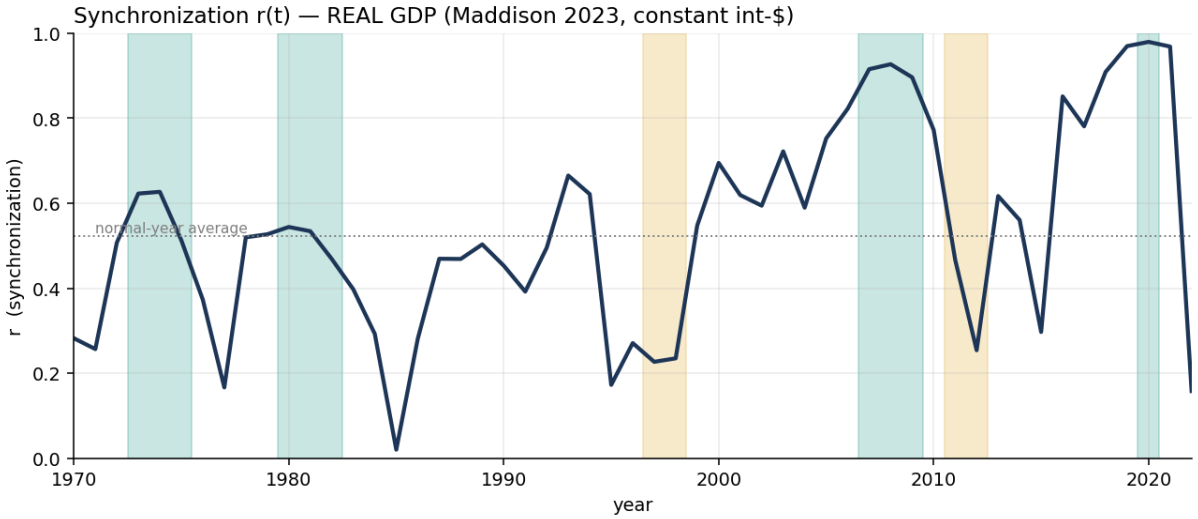


Figure 1. Real-GDP synchronization r(t), 1970–2022. Green = global crises, yellow = regional. The 2007–09 and 2020 windows stand out as sustained highs; regional crises sit at the troughs.

4 Zoom: each global crisis

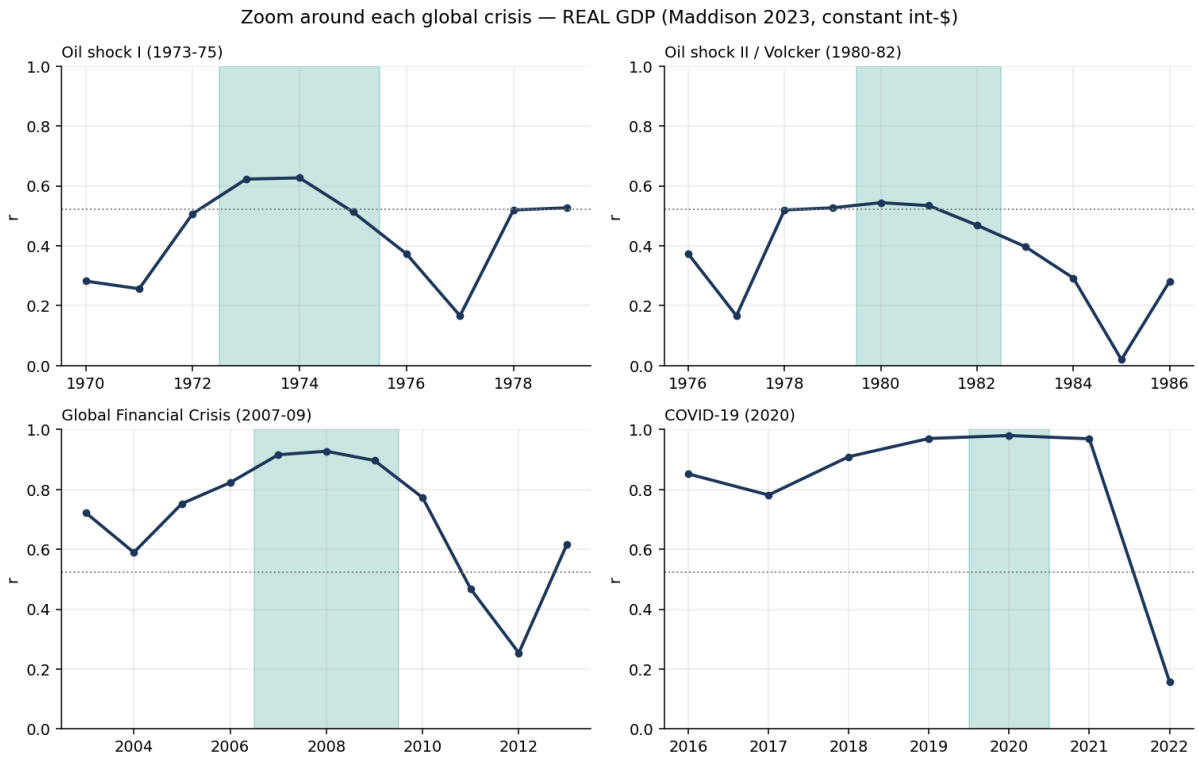


Figure 2. Real-GDP synchronization around each global crisis.

Global Financial Crisis (2007–09) is now the cleanest case: r climbs from the mid-2000s (0.75–0.82) to a sharp plateau of 0.92–0.90 (around the 92th percentile), then relaxes. **COVID-19 (2020)** is the most extreme point in the whole record: near-perfect synchronization (r = 0.97, 0.98, 0.97 across 2019–21) followed by an abrupt shattering to 0.16 in 2022 — textbook rise-then-rupture. The **oil shocks (1973–75, 1980–82)**, by contrast, are more muted here (r ≈ 0.5) than in current dollars: much of their apparent nominal synchronization was common global *inflation*, which real GDP removes. That is itself an informative result — the recent financial and pandemic crises are genuinely synchronized in real output, whereas the 1970s–80s oil episodes were partly a price-level phenomenon.

5 Composite, relative test, and group means

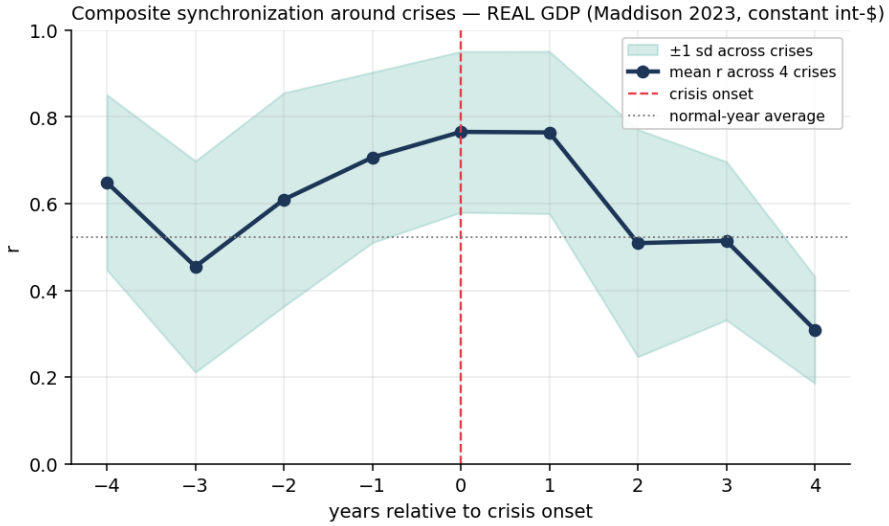


Figure 3. Composite (superposed-epoch) synchronization around global-crisis onsets, real GDP.

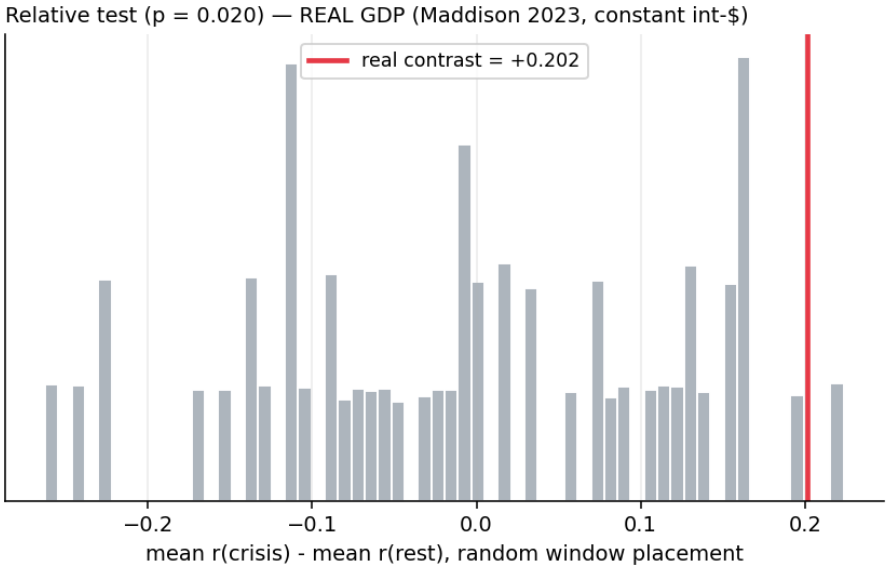


Figure 4. Relative test: the real global-crisis contrast (+0.202) versus random window placement gives p = 0.020 — stronger than the current-dollar run.

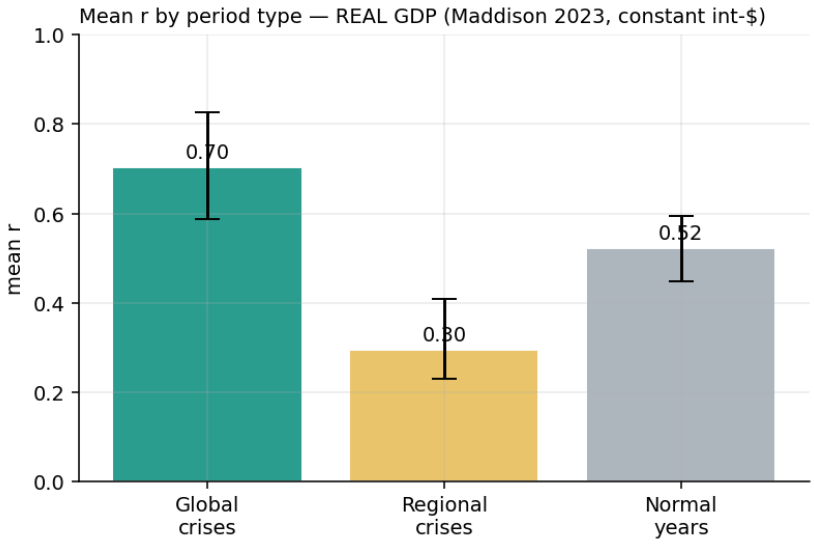


Figure 5. Mean r by period type (95% bootstrap CI): global 0.70 > normal 0.52 >> regional 0.30. The gaps are wider than in current dollars.

6 Current-dollar vs. real GDP: does it hold?

The headline numbers from both runs, side by side. The story is robust; real GDP sharpens it.

Quantity	Current US$	Real GDP
mean r, global crises	0.75	0.70
mean r, normal years	0.67	0.52
mean r, regional crises	0.45	0.30
global vs rest contrast	+0.103	+0.202
relative-test p-value	0.038	0.020
GFC 2008 (percentile)	68th	92th
COVID 2020 (percentile)	85th	98th

7 Key observations

- The core thesis is robust to the price basis: global crises are the most synchronized period type in both datasets, and the relative test is *stronger* in real GDP (p = 0.020).
- The 2007–09 and 2020 crises are unmistakable synchronization peaks in real terms — cleaner than in current dollars.
- COVID shows the sharpest fibrillation signature anywhere: near-perfect locking in 2019–21 (r up to 0.98) then a collapse to 0.16 in 2022.
- Regional crises are even more clearly the desynchronized troughs (percentiles in the single digits), confirming r separates global from regional shocks.
- The oil shocks look weaker in real GDP than in current dollars — evidence that part of their nominal ‘synchronization’ was shared global inflation rather than shared real contraction.

8 Limitations

- Real GDP per capita is annual and, for the modern period, still ultimately built on national-accounts growth; the 2020–22 window sits at the data edge (Maddison ends 2022), so the post-COVID collapse in 2022 should be read as indicative.
- Only four global crises and annual sampling limit statistical power, though the real-GDP signal is stronger than the nominal one.
- Per-capita real GDP removes population trends; using total real GDP would weight large economies differently (a minor robustness check).

9 Conclusion

Switching from current-dollar to real GDP does not weaken the economic-fibrillation result — it strengthens it. In constant prices, the world's economies lock together most tightly precisely during the truly global crises, most dramatically COVID-19, and fall apart both during regional shocks and in the immediate aftermath of the global ones. The one substantive change is that the 1970s–80s oil shocks lose prominence, revealing that their apparent synchronization was partly nominal. The core claim — high synchronization marks systemic fragility, and crises are desynchronization events —

survives the most demanding data check available here.